EXPERIMENT - 7

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Section/Group: KRG-2B
Semester: 5th
Date of Performance: 09/10/25
Subject Name: ADBMS
Subject Code: 23CSP-333

1. **AIM:**

i)Triggers: Student Data Change Monitoring (Medium)

EduSmart Institute wants to monitor all insertions and deletions in the student database. Whenever a new student record is inserted or deleted from the student table, the details of that record should be displayed on the PostgreSQL console window.

Objective:

Design a PostgreSQL trigger that:

1. Prints the complete details of the inserted or deleted student record using RAISE NOTICE.
2. Activates automatically after every INSERT or DELETE operation on the student table.

ii) Triggers: Employee Activity Logging (Hard)

TechSphere Solutions wants to maintain an automatic audit trail for all employee additions and deletions in the company database.

Whenever a new employee is added or removed from the tbl_employee table, an entry should be recorded in the tbl_employee_audit table for tracking purposes.

Objective:

Design a PostgreSQL trigger that:

1. Inserts a message in tbl_employee_audit whenever a new employee is added or deleted.
2. The message should include the employee's name and the current timestamp.
3. Activates automatically after every INSERT or DELETE operation on tbl_employee.

2. **Tools Used :** PostGres

Solutions:

Q1)

--CREATING A TABLE

```
CREATE TABLE student (
   id SERIAL PRIMARY KEY,
   name VARCHAR(100),
   age INT,
   class VARCHAR(50)
);


--TRIGGER FUNCTION

CREATE OR REPLACE FUNCTION fn_student_audit()
RETURNS TRIGGER
LANGUAGE plpgsql
AS
$$
BEGIN
   IF TG_OP = 'INSERT' THEN
      RAISE NOTICE 'Inserted Row -> ID: %, Name: %, Age: %, Class: %',
               NEW.id, NEW.name, NEW.age, NEW.class;
      RETURN NEW;

   ELSIF TG_OP = 'DELETE' THEN
      RAISE NOTICE 'Deleted Row -> ID: %, Name: %, Age: %, Class: %',
               OLD.id, OLD.name, OLD.age, OLD.class;
      RETURN OLD;
   END IF;

   RETURN NULL;
END;
$$;


--CREATING A TRIGGER
CREATE TRIGGER trg_student_audit
AFTER INSERT OR DELETE
ON student
FOR EACH ROW
EXECUTE FUNCTION fn_student_audit();
```

Q2)

```
CREATE TABLE tbl_employee (
   emp_id SERIAL PRIMARY KEY,
   emp_name VARCHAR(100),
   designation  VARCHAR(50),
   salary NUMERIC(10,2)
);
```

```
CREATE TABLE tbl_employee_audit (
   audit_id SERIAL PRIMARY KEY,
   message TEXT,
   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE OR REPLACE FUNCTION audit_employee_changes()
RETURNS TRIGGER
LANGUAGE plpgsql
AS
$$
BEGIN
   IF TG_OP = 'INSERT' THEN
      INSERT INTO tbl_employee_audit(message)
      VALUES ('Employee name ' || NEW.emp_name || ' has been added at ' || NOW());
      RETURN NEW;

   ELSIF TG_OP = 'DELETE' THEN
      INSERT INTO tbl_employee_audit(message)
      VALUES ('Employee name ' || OLD.emp_name || ' has been deleted at ' || NOW());
      RETURN OLD;
   END IF;

   RETURN NULL;
END;
$$;

CREATE TRIGGER trg_employee_audit
AFTER INSERT OR DELETE
ON tbl_employee
FOR EACH ROW
EXECUTE FUNCTION audit_employee_changes();


INSERT INTO tbl_employee (emp_name, designation, salary)
VALUES (‘Arpit Anand', 'Software Engineer', 55000);

SELECT * FROM tbl_employee_audit;

DELETE FROM tbl_employee WHERE emp_name = ‘Arpit Anand';

SELECT * FROM tbl_employee_audit;
```

3. **Output:**

query Query History Scratch Pad x

```
39  END;
40  ›s;
41
42
43  ——CREATING A TRIGGER
44  CREATE TRIGGER trg_student_audit
45  AFTER INSERT OR DELETE
46  ON student
47  FOR EACH ROW
08  EXECUTE FUNCTION fn_student_audit( ) ;

50  INSERT INTO student (name, age, class)
SI  VALUES ('Arpit Anand', 2 1, 'CS101');
52
```

Data Output Messages Notifications

```
NOTICE: Inserted Ro* -> ID: 1, Name: Arpit Anand, Age: 21, Class: CS101
INSfR7 0 i

Query returned successfully in 42 msec.
```

Query Query History Scratch Pad x

```
ou  on tbl_employee
54  FOR EACH ROU
s5  EXECUTE FUNCTION audit_employee_changes() ;
56

58  INSERT INTO tbl_employee (emp_name , designation , satary)
s9  VALUES ('Arpit Anand', 'Software Engineer', 55000) ;
60
61  SELECT x FROM tbl_employee_audit;
6 2
63  DELETE FROM tbl_employee WHERE emp_name = 'Arpit Anand';

65  SELECT * FROM tbt_emp1oyee_audit
66
```

Data Output Messages Notifications

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	audit_id [PK] integer	message text	created_at timestamp without time zone
1	1	Employee name Arpit Anand has been added at 2025-10-21 21:02:59.425952+0fi:30	2025-1 0-21 21:O2:59.425952

Query Query History Scratch Pad x

```
»u  Up to L_emp Loyee
s4  FOR EACH ROW
ss  EXECUTE FUNCTION audit_emptoyee_changes ( ) j
56
57
s8  INSERT INTO tbl_employee (emp_name , designation , salary)
s9  VALUES ('Arpit Anand', 'Software Engineer', 55000›;

61  SELECT FROM tbt_emptoyee_aud1 t;
62
63  DELETE FROM tbl_employee WHERE emp_name = 'Arpit Anand';

65  SELECT FROM tbt_employee_audit;
```

Data Output Messages Notifications

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	audit_id [PK] integer	message text	created_at timestamp without time zone
1	J	Employee name Arpit Anand has been added at 2025-10-21 21:02:59.425952+05.30	2025-1 0-21 21:02:59.425952
2	2	Employee name Arpit Anand has been deleted at 2025-10-21 21:03:19.998826*05:30	2025-1 0-21 21:03:J9.998826

4. Learning Outcomes:

1. Understand the concept and purpose of database triggers in PostgreSQL.
2. Learn how to automate data tracking using AFTER INSERT and AFTER DELETE triggers.
3. Gain hands-on experience with trigger functions written in PL/pgSQL.
4. Develop the ability to implement audit logging for real-time database monitoring.
5. Enhance skills in maintaining data integrity and traceability in relational databases.